

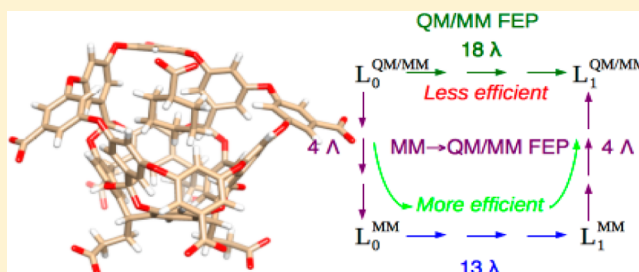
Comparison of QM/MM Methods To Obtain Ligand-Binding Free Energies

Martin A. Olsson and Ulf Ryde*

Department of Theoretical Chemistry, Lund University, Chemical Centre, P.O. Box 124, SE-221 00 Lund, Sweden

Supporting Information

ABSTRACT: We have compared two approaches to calculate relative binding free energies employing molecular dynamics simulations at the combined quantum-mechanical/molecular mechanics (QM/MM) level. As a test case, we study the binding of nine cyclic carboxylate ligands to the octa-acid deep-cavitand host system. The ligand is treated with the semiempirical PM6-DH+ QM method. In the first approach, we perform direct alchemical QM/MM free energy perturbation (FEP). In the second, reference-potential approach, we convert the ligands with FEP at the molecular mechanics (MM) level and then perform also MM $\rightarrow$ QM/MM FEP for each ligand. We show that the two approaches give identical results within statistical uncertainty. For the reference-potential approach, the MM $\rightarrow$ QM/MM perturbation converges in terms of energies, uncertainties, and overlap measures with two intermediate states, giving a precision of 0.5–0.9 kJ/mol for all eight transformations considered. On the other hand, the QM/MM-FEP approach requires 17–18 intermediate states, showing that the reference-potential approach is more effective. Previous calculations with single-step exponential averaging (i.e., entirely avoiding QM/MM simulations) required fewer QM/MM energy calculations, but they gave worse precision and involved approximations with an unclear effect on the results.



INTRODUCTION

One of the largest challenges of computational chemistry is to predict the binding affinities of small molecules to biological macromolecules. If such affinities could reliably be predicted with a high accuracy for any type of system, this would revolutionize drug discovery and development. Consequently, numerous methods have been suggested with this aim.^{1,2} The most widely used approach is docking with simple scoring functions based on either a statistical survey of structures with known affinities or physical energy functions.^{3,4} At the next step of approximation come approaches based on molecular dynamics (MD) simulations of the physical end states (the complex and possibly also the free ligand and the free macromolecule), e.g., molecular mechanics combined with Poisson–Boltzmann and surface area solvation (MM/PBSA), linear interaction energy (LIE), and similar approaches.^{5–8} However, from a statistical mechanics perspective, the proper way to obtain binding free energies is by the use of free energy perturbation (FEP) methods based on MD or Monte Carlo simulations of both the end states and various intermediate states and calculating the free energies by methods such as exponential averaging (EA) (Zwanzig equation), thermodynamic integration (TI), or the (multistate) Bennett acceptance ratio (BAR) approach.^{9–14} Recently, several applications of FEP to calculate relative binding free energies for large sets of protein–ligand complexes have shown a promising accuracy, at least for some proteins.^{15–17} FEP methods can also be used to calculate absolute affinities by converting the ligand to a noninteracting molecule.^{18–20}

FEP simulations should in principle give the correct results provided that the energy function is perfect and the sampling is infinite. Consequently, there has recently been quite some interest in improving FEP calculations of binding affinities by the use of quantum-mechanical (QM) calculations instead of the normal molecular mechanics (MM) force field.²¹ Several different approaches have been employed. If only the ligand is treated with a cheap semiempirical QM method, a full FEP can be performed with the combined QM/MM potential. Only a few applications with such an approach have been presented, but often with impressive results.^{22–24}

However, it has been more common to employ more accurate QM methods (e.g., density functional theory or coupled-cluster calculations) and reference-potential methods.^{25–27} In these, the phase space is still sampled with MM methods, and these simulations are then used to perform an FEP in the method space from the MM to the QM/MM potential. This perturbation can in principle be performed for all intermediate states between the two ligands,^{28,29} but it is more common to perform it only for the end states (i.e., the two ligands).^{30,31} Then, the difference in binding energy between the two ligands is estimated by FEP at the MM level, and this energy is corrected toward the QM/MM potential by one MM $\rightarrow$ QM/MM FEP for each of the ligands. The MM $\rightarrow$ QM/MM free energy can be obtained either by reweighting the MM snapshots, e.g., by the non-Boltzmann BAR

Received: December 15, 2016

Published: March 29, 2017

approach (NBB)³² or by employing single-step EA (ssEA) from MM to QM/MM.^{25,28,29} The two approaches have been compared for the same system, and it has been shown that ssEA gives the more precise results both analytically and practically and actually also with a lower computational effort.^{29,33,34} Alternative approaches employ energy distribution functions.³⁵

The prime problem with these approaches is to obtain a proper convergence of the MM $\rightarrow$ QM/MM perturbation. For normal FEP simulations, the perturbation is divided into many small steps in order to ensure a proper overlap between the two energy distributions. However, to avoid doing the extremely time-consuming sampling at the QM/MM level, the MM $\rightarrow$ QM/MM perturbation needs to be done in a single step (employing sampling only at the MM level). If only the ligand is treated by QM or only electrostatic interactions are considered, apparently converged results have been obtained in a few cases.^{30,36} However, if also the closest parts of the receptor are included in the QM region, poor precision has been observed with NBB and ssEA approaches when performing some thousands of QM calculations for each state.^{28,31} Recently, we showed that 720 000 QM calculations are needed to converge the MM $\rightarrow$ QM/MM perturbation to within 1 kJ/mol for the relative binding affinities of a series of benzoate ligands to the octa-acid host–guest system.²⁹ Moreover, it was essential to employ a cumulant approximation^{37,38} for the calculation of the ssEA free energies (i.e., assuming that the energies follow a Gaussian distribution).

Performing 720 000 QM evaluations corresponds to carrying out a fairly long MD simulation (1.4 ns with a 2 fs time step). The question therefore naturally arises whether it is better to actually perform QM/MM MD simulations to improve the convergence of the MM $\rightarrow$ QM/MM perturbation or perhaps to perform the full FEP at the QM/MM level. The advantage of such an approach is that it avoids reliance on the cumulant approximation, which is hard to confirm for this type of system. In fact, even after 720 000 QM evaluations, there were some indications that the estimate of the precision was still too optimistic, with variations of up to 9 kJ/mol in some of the free energy components among independent simulations.²⁹ Moreover, total energies can be employed, whereas most MM $\rightarrow$ QM/MM perturbations have used instead interaction energies, excluding differences in the internal energy of the QM region.^{28,29,31,33}

QM/MM sampling has been used in a few reference-potential studies of ligand binding. Mulholland and co-workers used QM/MM Monte Carlo sampling speeded up by the Metropolis–Hastings approach³⁹ to study the binding of water molecules to two sites in neuraminidase (with only the water molecule treated by QM).⁴⁰ Such an approach required 160 000 QM evaluations for each state, and they observed differences of up to 12 kJ/mol between the MM and QM descriptions of the water molecules. Moreover, Skylaris and co-workers employed QM/MM simulations for the end points and one intermediate state as a stepping stone between MM and full QM calculations of the hydration energy of five small organic molecules.³⁶ Converged results were obtained with only 500 QM and QM/MM evaluations for each step. Boresch and co-workers showed that the MM $\rightarrow$ QM free energy can be estimated from short QM simulations (50–500 time steps but repeated 1000 times) employing nonequilibrium work functions.⁴¹ For enzyme reactions, Warshel and co-workers have suggested the par-dynamics approach, which employs the reference-potential method but improves the convergence of the free energies by also performing the QM/MM MD simulation at the end states

and calculating the free energy with the linear-response approximation.^{25,42,43} However, recent studies have shown that this approximation sometimes gives poor results compared to a full FEP calculation at the QM/MM level (errors of up to 92 kJ/mol),⁴⁴ and it is not clear how it could be known beforehand whether the approximation will work (i.e., without performing the full QM/MM FEP).

In this work, we have performed calculations of relative binding free energies based on explicit QM/MM MD simulations. We investigate the convergence of reference-potential FEP calculations with up to seven intermediate states in the MM $\rightarrow$ QM/MM perturbations, discussing the accuracy of the linear-response approximation and how it can be estimated without performing additional simulations. Moreover, we compare the accuracy and efficiency of reference-potential calculations with full QM/MM FEP simulations of the same systems. As a test system, we employ the binding of nine small carboxylate ligands to the octa-acid deep cavitand,^{45–47} which was employed in the SAMPL4 competition^{48,49} and was also used in several of our previous studies.^{28,29,50} To speed up the calculations and make it possible to compare them to full QM/MM FEP, we employed the semiempirical PM6-DH+ approach for the QM calculations,^{51–53} including only the ligand in the QM system.

METHODS

Simulated System. We have studied the relative binding free energies of nine cyclic carboxylate ligands to the octa-acid deep-cavity host, which was involved in the SAMPL4 competition.^{45–49} The octa-acid host and the nine ligands are shown in Figure 1. The MM calculations are taken from our previous studies of this system.^{28,29} They employed the **general Amber force field** for both the octa-acid host and the guest molecules,⁵⁴ using **symmetrized restrained electrostatic potential charges**⁵⁵ obtained at the **Hartree–Fock/6-31G*** level of theory.²⁸ The **TIP3P** model was used to describe the water molecules.⁵⁶ As in our previous studies,^{28,29} eight ligand transformations were considered: MeBz $\rightarrow$ Bz, EtBz $\rightarrow$ MeBz, pClBz $\rightarrow$ Bz, mClBz $\rightarrow$ Bz, Hx $\rightarrow$ Bz, MeHx $\rightarrow$ Hx, Hx $\rightarrow$ Pen, and Hep $\rightarrow$ Hx (employing the ligand names defined in Figure 1).

QM/MM Free Energy Simulations. Here we calculate $\Delta\Delta G_{L_0 \rightarrow L_1}^{\text{bind}}$, the difference between the binding free energies of pairs of ligands (L_0 and L_1) to the octa-acid host, by converting L_0 to L_1 both when the ligands are bound to the host ($\Delta G_{L_0 \rightarrow L_1}^{\text{bound}}$) and when they are free in solution ($\Delta G_{L_0 \rightarrow L_1}^{\text{free}}$):⁵⁷

$$\Delta\Delta G_{L_0 \rightarrow L_1}^{\text{bind}} = \Delta G_{L_0 \rightarrow L_1}^{\text{bound}} - \Delta G_{L_0 \rightarrow L_1}^{\text{free}} \quad (1)$$

Each term on the right-hand side of this equation was estimated by FEP, slowly converting L_0 to L_1 , using the energy function

$$E(\lambda) = (1 - \lambda)E_0 + \lambda E_1 \quad (2)$$

where E_0 and E_1 are the potentials of the two ligands and λ is a coupling parameter that goes from 0 to 1. Our aim is to calculate each term $\Delta G_{L_0 \rightarrow L_1}^s$ (where $s = \text{bound or free}$) at the QM/MM level. We will compare two approaches, as illustrated in Figure 2. In the first, all of the MD simulations at different λ values were performed at the QM/MM level. This is illustrated by the upper horizontal part of the thermodynamic cycle in Figure 2. We will call this approach QM/MM-FEP.

In the second approach, we tried to decrease the number of demanding QM/MM MD simulations by employing the full

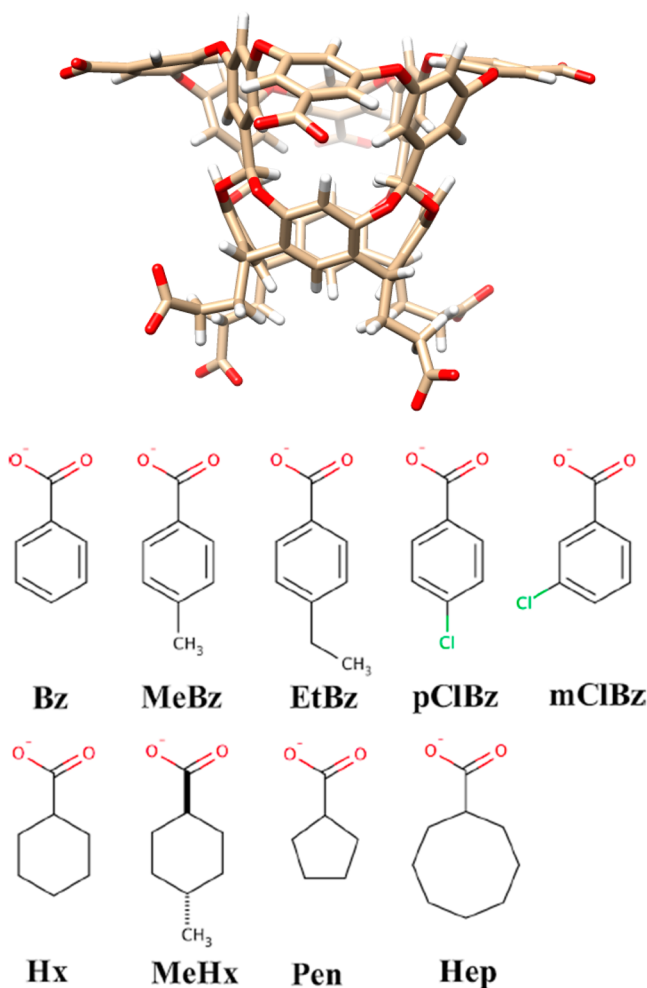


Figure 1. Octa-acid host and the nine guest molecules considered in this study.

thermodynamic cycle in Figure 2. Thus, the MD simulations at different λ values were performed at the MM level, giving $\Delta\Delta G_{L_0 \rightarrow L_1}^{s,MM}$ (lower horizontal path in Figure 2). These values were calculated by employing 13 λ values, viz., 0.0, 0.05, 0.1, 0.2, ..., 0.8, 0.9, 0.95, and 1.0. Then at the end states (L_0 and L_1) we performed additional FEP calculations in the method space, going from an MM description to a QM/MM description ($\Delta\Delta G_{L_i}^{s,MM \rightarrow QM/MM}$ for $i = 0, 1$). On the basis of the thermodynamic cycle in Figure 2, the net QM/MM free energy can be estimated from these three paths:

$$\Delta\Delta G_{L_0 \rightarrow L_1}^{s,QM/MM} = \Delta\Delta G_{L_0 \rightarrow L_1}^{s,MM} - \Delta\Delta G_{L_0}^{s,MM \rightarrow QM/MM} + \Delta\Delta G_{L_1}^{s,MM \rightarrow QM/MM} \quad (3)$$

In previous studies, we attempted to avoid any QM/MM MD simulations by calculating the free energy differences $\Delta\Delta G_{L_i}^{s,MM \rightarrow QM/MM}$ ($i = 0, 1$) on the basis of only a single MD simulation at the MM level using the ssEA approach.^{28,29,31} However, as discussed in the Introduction, this led to severe convergence problems, requiring 720 000 QM calculations for proper convergence.²⁹ Therefore, we here study whether the convergence of the MM $\rightarrow$ QM/MM perturbation can be improved by performing simulations with intermediate states. In analogy with eq 2, we form the energy function

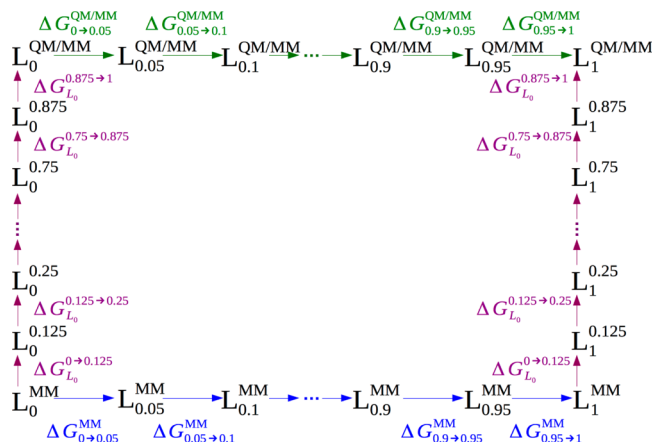


Figure 2. Thermodynamic cycle employed to study the relative free energy for the binding of two ligands (L_0 and L_1) to a target receptor. The cycle applies both when the ligand is bound to the receptor and when it is free in solution. We aim to estimate the free energy difference for converting L_0 to L_1 at the QM/MM level, either by directly calculating it by QM/MM MD simulations at a number of different λ values (top horizontal reaction with green arrows) or by converting L_0 to L_1 at the MM level (bottom horizontal reaction, blue arrows) combined with MM $\rightarrow$ QM/MM perturbations for each of the two ligands (left and right vertical reactions, magenta arrows), again by employing a number of intermediate Λ states.

$$E(\Lambda) = (1 - \Lambda)E_{MM} + \Lambda E_{QM/MM} \quad (4)$$

where E_{MM} is the MM energy, $E_{QM/MM}$ is the QM/MM energy, and Λ is the coupling parameter (we use the symbol Λ here to distinguish it from the λ parameter used to convert L_0 to L_1 in eq 2). In previous studies, we performed MD simulations only for $\Lambda = 0$, whereas in this study we use two, three, five, or nine Λ values at regular intervals (i.e., $\Lambda = 0.0, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 0.875$, and 1.0 for nine Λ values and a subset of these values for the other three sets). In addition, we also tried four Λ values at $\Lambda = 0.0, 0.333, 0.666$, and 1.0.

Details of the Calculations. All of the FEP and MD simulations were performed with the sander module of the Amber 14 software.⁵⁸ The QM/MM calculations employed the QM/MM implementation in this software^{58,59} with electrostatic embedding. For the explicit QM/MM FEP simulations (top reaction in Figure 2), we used a dual-topology scheme with the energy function

$$E^{QM/MM}(\lambda) = (1 - \lambda)E_0^{QM/MM} + \lambda E_1^{QM/MM} \quad (5)$$

Thus, for each λ value, two separate QM calculations were performed, one for L_0 and one for L_1 . Then the energies were combined using eq 5, which shows how energies for the mixed states ($0 < \lambda < 1$) are naturally obtained.

To speed up the calculations and make full QM/MM FEP calculations possible, only the guest molecule was included in the QM region. The QM calculations were performed at the semiempirical PM6-DH+ level of theory.^{51–53} The QM/MM FEP simulations were run on 20 CPU cores at a speed of 1–5 ns/day.

Free energy differences were calculated with the multistate Bennett acceptance ratio (MBAR) approach using the pyMBAR software.⁶⁰ This software also gives uncertainties of the calculated free energies and selects a noncorrelated subset of data points. For comparison and to evaluate the convergence, the

Table 1. $\Delta\Delta G_L^{MM \rightarrow QM/MM}$ QM/MM Free Energy Corrections (in kJ/mol) for the Eight Transformations Calculated with Two, Three, Four, Five, or Nine Λ Values for Both L_0 and L_1 ; The Last Row Shows the Mean Absolute Deviation (MAD) from the Calculations with Nine Λ Values

	ligand	two Λ values	three Λ values	four Λ values	five Λ values	nine Λ values
MeBz $\rightarrow$ Bz	MeBz	1.9 ± 4.6	-0.9 ± 1.0	-0.6 ± 0.4	-0.9 ± 0.7	-1.3 ± 0.5
	Bz	-1.5 ± 5.0	0.0 ± 0.9	-0.3 ± 0.4	-0.4 ± 0.6	-1.0 ± 0.5
EtBz $\rightarrow$ MeBz	EtBz	1.3 ± 4.9	2.1 ± 1.0	1.9 ± 0.4	2.0 ± 0.7	2.1 ± 0.5
	MeBz	3.0 ± 3.3	1.1 ± 1.0	-0.3 ± 0.4	-0.8 ± 0.6	-0.8 ± 0.5
pClBz $\rightarrow$ Bz	pClBz	-3.4 ± 2.6	-5.2 ± 0.9	-5.0 ± 0.3	-4.8 ± 0.6	-5.5 ± 0.5
	Bz	-4.5 ± 7.2	-1.4 ± 1.0	-0.3 ± 0.4	-1.1 ± 0.6	-0.9 ± 0.5
mClBz $\rightarrow$ Bz	mClBz	-13.5 ± 4.7	-4.2 ± 1.0	-4.1 ± 0.4	-3.6 ± 0.7	-3.8 ± 0.5
	Bz	-3.3 ± 3.4	-1.4 ± 1.0	-0.8 ± 0.4	-0.6 ± 0.6	-1.0 ± 0.5
Hx $\rightarrow$ Bz	Hx	1.4 ± 2.4	2.3 ± 0.9	2.9 ± 0.3	2.8 ± 0.6	2.9 ± 0.4
	Bz	-0.6 ± 2.3	-0.9 ± 0.9	-0.7 ± 0.4	-1.4 ± 0.6	-0.8 ± 0.5
MeHx $\rightarrow$ Hx	MeHx	-2.9 ± 2.2	-0.8 ± 0.9	-0.6 ± 0.3	-1.1 ± 0.6	-0.6 ± 0.4
	Hx	3.7 ± 2.5	2.3 ± 0.8	2.1 ± 0.4	3.0 ± 0.6	2.7 ± 0.4
Hx $\rightarrow$ Pen	Pen	4.0 ± 3.2	3.4 ± 0.9	2.9 ± 0.3	3.1 ± 0.6	2.4 ± 0.5
	Hx	0.7 ± 2.3	2.3 ± 0.9	3.3 ± 0.3	2.0 ± 0.6	2.7 ± 0.4
Hep $\rightarrow$ Hx	Hep	-4.0 ± 3.0	1.8 ± 0.9	2.1 ± 0.4	0.6 ± 0.6	0.9 ± 0.5
	Hx	7.0 ± 2.4	3.5 ± 0.9	3.6 ± 0.3	3.0 ± 0.6	3.0 ± 0.4
MAD		2.7	0.5	0.4	0.4	

free energies were calculated also with BAR, TI, and EA using the same software.^{10–12}

The octa-acid host and the guest molecules were solvated in a truncated octahedral box of explicit water molecules extending at least 10 Å from the solute using the leap program in the Amber suite, giving ~ 5000 and ~ 1700 atoms in total for the calculations with and without the host, respectively. No counterions were employed—our previous test calculations indicated that they have only a small effect on the relative binding free energies (< 2 kJ/mol).²⁸

We first performed 500 cycles of minimization with the heavy atoms of the host and guest molecules restrained toward the starting structure with a force constant of $418.4 \text{ kJ mol}^{-1} \text{ \AA}^{-2}$. This was followed by 20 ps of constant-volume equilibration with the same restraints and 0.5 ns of constant-pressure equilibration without any restraints. Finally, a 1 ns production simulation was run, during which energies were sampled every 5 ps (every 1 ps for the simulations with four Λ values).

The temperature was kept constant at 300 K using Langevin dynamics with a collision frequency of 2 ps^{-1} .⁶¹ The pressure was kept constant at 1 atm using Berendsen's weak-coupling isotropic algorithm with a relaxation time of 1 ps.⁶² Long-range electrostatics were handled by particle-mesh Ewald summation⁶³ with a fourth-order B-spline interpolation and a tolerance of 10^{-5} . The cutoff radius for Lennard–Jones interactions was set to 8 Å. The MM calculations were performed with all bonds involving hydrogen atoms constrained to their equilibrium values using the SHAKE algorithm,⁶⁴ allowing for a time step of 2 fs during the simulations. However, this was not allowed within the QM region. Therefore, all of the QM/MM MD simulations were run without any constrained bonds and a time step of 1 fs. Test QM/MM-FEP simulations showed that a time step of 0.5 fs gave identical results (within 0.1 kJ/mol for the pClBz $\rightarrow$ Bz transformation).

Quality Measures. To study the convergence for the various perturbations, several overlap measures were employed, viz., the Bhattacharyya coefficient (Ω),⁶⁵ the Wu and Kofke overlap measures of the energy probability distributions (K_{AB}),⁶⁶ and their bias metrics (Π).⁶⁶ The quality of the FEP calculations was further analyzed using the weight of the maximum term in the

exponential average ($w_{\max}$),²⁷ the difference of the forward and backward exponential average estimates ($\Delta\Delta G_{EA}$), and the difference between the BAR and TI estimates ($\Delta\Delta G_{TI}$).¹⁶

The performance of the free energy estimates compared with experimental data was quantified by the mean absolute deviation (MAD), the correlation coefficient (R^2), and Kendall's rank correlation coefficient (τ_r). The last of these was calculated only for the transformations that were explicitly studied, not for all combinations that could be formed from these transformations. The standard deviations of the quality measures were obtained by a simple simulation approach.⁶⁷ Each transformation was assigned a random number from a Gaussian distribution with the mean and standard deviation of the mean obtained from the BAR calculations. The quality measures were then calculated, and the procedure was repeated 1000 times.

RESULTS AND DISCUSSION

In this work, we have compared two methods to estimate the difference in binding free energy of two ligands to a receptor at the QM/MM level of theory using FEP calculations. As a test system, we have used the binding of nine cyclic carboxylate ligands to the octa-acid deep-cavity host.⁴⁰ The experimental data were taken from the SAMPL4 challenge.^{48,49} The octa-acid host is a macrocycle of 184 atoms with fourfold symmetry, 12 benzene rings, and eight carboxylic groups (Figure 1). It forms a hydrophobic cavity with a depth of 10 Å that can bind various small molecules in aqueous solution. The nine ligands are also shown in Figure 1. They are all cyclic (with five to seven atoms in the ring) with a carboxylate group and therefore a single negative charge. We studied eight ligand transformations: MeBz $\rightarrow$ Bz, EtBz $\rightarrow$ MeBz, pClBz $\rightarrow$ Bz, mClBz $\rightarrow$ Bz, Hx $\rightarrow$ Bz, MeHx $\rightarrow$ Hx, Hx $\rightarrow$ Pen, and Hep $\rightarrow$ Hx. These mostly involve H $\rightarrow$ methyl or H $\rightarrow$ Cl substitutions, although two involve the deletion of a CH_2 group inside the ring (Hx $\rightarrow$ Pen and Hep $\rightarrow$ Hx) and one involves the conversion of cyclohexane to benzene (Hx $\rightarrow$ Bz). We have performed two types of FEP calculations, which are described in separate sections.

Reference-Potential Calculations. We first tried to calculate the relative binding free energies by the reference-potential approach using the full thermodynamic cycle in Figure

2. Thus, we first calculated $\Delta\Delta G_{L_0 \rightarrow L_i}^{MM}$ at the MM level (lower horizontal branch in Figure 2). This involved normal FEP calculations at the MM level, and these results were taken from our previous MM study of the same system and will not be much discussed.²⁸ Then we calculated the free energies for conversion of each of the two ligands from an MM description to a QM/MM description, both when the ligands are bound to the host or when they are free in solution ($\Delta\Delta G_{L_i}^{s,MM \rightarrow QM/MM}$ for $L_i = L_0$ or L_1 and $s = \text{bound or free}$; i.e., the two vertical branches in Figure 2).

In previous studies, we and others have tried to estimate this term with ssEA or NBB, thereby avoiding MD simulations at the QM/MM level.^{28–31,33} However, it has been observed that the convergence is very poor, requiring at least 720 000 QM calculations.²⁹ Here we investigate whether the convergence can be improved by performing QM/MM MD simulations at the end points and possibly at additional intermediate states, described by the Λ coupling parameter in eq 4. We have tested calculations with two, three, four, five, and nine Λ values.

The individual $\Delta\Delta G_L^{s,MM \rightarrow QM/MM}$ terms were rather large, ranging from -478 to -768 kJ/mol (Table S1). However, for each ligand, the values were of a similar size when the ligand was bound to the host or free in water solution. Therefore, the difference between these two calculations, $\Delta\Delta G_L^{MM \rightarrow QM/MM} = \Delta G_L^{\text{bound},MM \rightarrow QM/MM} - \Delta G_L^{\text{free},MM \rightarrow QM/MM}$, which is shown in Table 1, was much smaller (e.g., -5.5 to 3 kJ/mol with nine Λ values).

From the results in Table 1, it can directly be seen that the calculations with two Λ values gave unreliable results. The calculated $\Delta\Delta G_L^{MM \rightarrow QM/MM}$ values differed by up to 10 kJ/mol from the results obtained with nine Λ values (for the individual $\Delta G_L^{s,MM \rightarrow QM/MM}$ results in Table S1, the difference was somewhat smaller, up to 6 kJ/mol). The calculations with three, four, and five Λ values gave appreciably smaller differences, up to 1.8 , 1.5 , and 0.8 kJ/mol, respectively, with average differences of 0.5 , 0.4 , and 0.4 kJ/mol, respectively.

Likewise, the standard errors of the calculated $\Delta\Delta G_L^{MM \rightarrow QM/MM}$ free energies were also large and varying with two Λ values ($2\text{--}7$ kJ/mol). With more Λ values, the precision improved strongly: already with three Λ values it was $0.8\text{--}1.0$ kJ/mol for all of the calculations, and it decreased to $0.4\text{--}0.5$ kJ/mol for nine Λ values. The precision with four Λ values was even better (0.3 kJ/mol) because energy data were collected five times more often (every 1 ps instead of every 5 ps). The free energies were obtained from a noncorrelated subset of the calculated energies, estimated using the pyMBAR software.⁶⁰ However, omission of correlated data changed the resulting free energies by less than 0.8 kJ/mol, indicating that the sampling frequency of 5 ps in the other calculations was unnecessarily sparse (the correlation time of the potential energy differences estimated by pyMBAR was $1\text{--}6$ ps, with averages of 1.6 and 1.2 ps for the bound and free simulations, respectively).

These results were obtained with the MBAR method. Quite satisfyingly, BAR calculations gave essentially identical results, with an average deviation of only 0.04 kJ/mol (maximum difference 0.2 kJ/mol). However, TI and EA calculations gave larger deviations of 0.7 and 2.9 kJ/mol on average (maximum 8 and 24 kJ/mol), respectively. The difference was largest for the calculations with two Λ values, 2.7 and 6.7 kJ/mol on average, and decreased to 0.1 and 0.3 kJ/mol for nine Λ values. This reflects the poorer overlap with fewer Λ values, which is partly amended by the use of BAR or MBAR.

Further estimates of the convergence can be obtained by observing the fact that the $\Delta\Delta G_L^{s,MM \rightarrow QM/MM}$ free energies depend only on a single ligand (i.e., they do not depend on the other ligand in the perturbation). This means that we actually have four independent estimates of $\Delta\Delta G_L^{s,MM \rightarrow QM/MM}$ for the Bz and Hx ligands and two estimates for the MeBz ligand in Table S1. For the individual calculations with two Λ values, the results varied by up to 6.6 kJ/mol with a standard deviation of 2.0 kJ/mol. With three, four, five, and nine Λ values, the standard deviation decreased to 0.6 , 0.5 , 0.4 , and 0.2 kJ/mol, respectively, and the maximum deviation was 2.6 , 2.5 , 1.3 , and 0.8 kJ/mol, respectively. The results were similar when taking the difference of the calculations with the ligand bound to the host or free in solution, as shown in Table 1 (e.g., the standard deviation was 1.9 , 0.7 , 0.7 , 0.4 , and 0.1 kJ/mol with two, three, four, five, and nine Λ values, respectively).

Thus, the results in Tables 1 and S1 show that two Λ values are too few to obtain reliable results for the $MM \rightarrow QM/MM$ perturbation; instead, three to five Λ values should be used, depending on the desired accuracy and precision. However, it is likely that this conclusion may depend on what system is studied. Therefore, it would be strongly desirable if we could decide directly from the calculations with a certain number of Λ values whether the results are reliable. We have seen that the standard error of the calculated free energies give a first indication in this direction.

To obtain more quantitative measures, we tested six overlap measures that have been employed to judge the overlap in previous FEP calculations,^{16,29} viz., the Bhattacharyya coefficient (Ω),⁶⁵ the Wu and Kofke overlap measures (K_{AB})⁶⁶ and their bias metrics (Π),⁶⁶ the weight of the maximum term in the exponential average ($w_{\max}$),²⁷ the difference of the forward and backward exponential average estimates ($\Delta\Delta G_{EA}$), and the difference between the BAR and TI estimates ($\Delta\Delta G_{TI}$).¹⁶ To start with, we used the following indicators of poor overlap: $\Omega < 0.7$, $K_{AB} < 0.7$, $\Pi < 0.5$, $w_{\max} > 0.3$, $\Delta\Delta G_{EA} > 4$ kJ/mol, and $\Delta\Delta G_{TI} > 4$ kJ/mol.^{16,66} The results were quite clear and consistent: For the calculations with two and three Λ values, several measures indicated poor overlap, in particular Π and $w_{\max}$ and in most cases also $\Delta\Delta G_{EA}$. On the other hand, most of the criteria indicated proper overlap for the calculations with four, five, and nine Λ values. The only exception was $w_{\max}$, which sometimes indicated a poor overlap for the calculations with four or five Λ values.

Except for the Π measure,⁶⁶ the overlap criteria used are rather arbitrary, without any theoretical basis. Considering that the results were so consistent, we decided to determine improved limits for the various overlap measures based on these results, optimizing the consistency with the results of the Π measure, i.e., that all calculations with two and three Λ values show poor overlap whereas those with four, five, and nine Λ values have acceptable overlap. For Ω , the best results were obtained between 0.84 and 0.85 , for which only three of the 64 calculations gave results that were not consistent with the Π measure. K_{AB} attains values between 0 and 2 , and in the present study they are always larger than unity. A double-sided criterion seems to be necessary. In the present study, values between 1.25 and 1.27 gave only three calculations that disagreed with the Π measure. Thus, $|K_{AB} - 1| > 0.25$ seemed to be a proper criterion, also agreeing with Ω . For $w_{\max}$ the original criterion was apparently too low. Instead, the best agreement with the Π measure was obtained between 0.47 and 0.60 , for which only three

Table 2. Overlap Measures for MM $\rightarrow$ QM/MM Perturbations of the Various Transformations for Various Numbers of Λ Values (2–9)^a

	no. of Λ values	L_0						L_1					
		Ω	K_{AB}	Π	w_{max}	$\Delta\Delta G_{EA}$	$\Delta\Delta G_{TI}$	Ω	K_{AB}	Π	w_{max}	$\Delta\Delta G_{EA}$	$\Delta\Delta G_{TI}$
MeBz $\rightarrow$ Bz	2	0.4	1.7	−3.0	1.00	14.3	0.1	0.4	1.7	−2.3	0.99	35.0	5.0
	3	0.7	1.3	−0.8	0.85	3.7	0.2	0.7	1.3	−0.5	0.68	5.9	0.3
	4	0.9	1.3	0.7	0.23	0.8	0.0	0.9	1.3	0.8	0.32	0.4	0.0
	5	0.9	1.1	0.5	0.29	0.8	0.1	0.9	1.1	0.9	0.20	1.1	0.2
	9	0.9	1.1	1.4	0.06	0.4	0.0	0.9	1.0	1.4	0.06	0.5	0.1
EtBz $\rightarrow$ MeBz	2	0.7	1.7	−2.1	0.93	35.0	3.2	0.6	1.7	−2.4	1.00	22.8	1.5
	3	0.9	1.4	−0.3	0.62	3.8	0.9	0.9	1.3	−0.4	0.82	4.5	0.3
	4	0.9	1.3	0.5	0.68	3.9	0.0	1.0	1.2	0.8	0.24	0.9	0.0
	5	0.9	1.1	0.9	0.17	1.0	0.3	0.9	1.1	0.9	0.23	1.3	0.1
	9	0.9	1.0	1.6	0.04	0.4	0.1	0.9	1.1	1.3	0.16	0.5	0.0
pClBz $\rightarrow$ Bz	2	0.7	1.7	−2.3	0.87	23.4	0.1	0.6	1.6	−2.4	1.00	39.9	4.1
	3	0.8	1.3	−0.5	0.72	3.2	0.6	0.8	1.3	−0.4	0.81	7.2	0.3
	4	1.0	1.3	0.8	0.17	1.1	0.0	1.0	1.2	1.0	0.12	1.5	0.0
	5	0.9	1.1	0.7	0.23	1.1	0.1	0.9	1.1	0.9	0.21	1.4	0.2
	9	0.9	1.0	1.4	0.08	0.7	0.0	0.9	1.1	1.6	0.08	0.5	0.0
mClBz $\rightarrow$ Bz	2	0.6	1.7	−2.3	1.00	34.8	8.3	0.7	1.6	−2.3	0.94	26.9	3.0
	3	0.8	1.3	−0.7	0.92	5.0	0.5	0.8	1.3	−0.7	0.93	7.0	0.6
	4	1.0	1.2	0.8	0.25	1.5	0.0	1.0	1.2	0.8	0.15	1.0	0.0
	5	0.9	1.1	0.8	0.35	1.8	0.1	0.9	1.1	0.8	0.49	1.5	0.0
	9	0.9	1.1	1.4	0.15	0.5	0.1	0.9	1.0	1.6	0.08	0.4	0.0
Hx $\rightarrow$ Bz	2	0.5	1.6	−2.3	0.99	18.7	1.9	0.4	1.6	−2.9	1.00	13.0	0.7
	3	0.8	1.3	−0.6	0.87	3.3	0.1	0.7	1.3	−0.8	0.91	3.4	0.8
	4	0.9	1.2	0.8	0.22	1.1	0.0	0.9	1.2	0.6	0.22	0.9	0.0
	5	0.9	1.1	0.6	0.33	1.4	0.0	0.9	1.1	0.6	0.38	1.0	0.1
	9	0.9	1.0	1.5	0.06	0.5	0.0	0.9	1.0	1.5	0.08	0.6	0.1
MeHx $\rightarrow$ Hx	2	0.7	1.6	−2.0	0.74	17.3	3.3	0.6	1.6	−1.8	0.83	19.8	1.9
	3	0.9	1.3	−0.3	0.81	2.2	0.4	0.8	1.2	−0.3	0.43	4.3	0.7
	4	1.0	1.2	0.8	0.24	1.0	0.0	0.9	1.1	0.9	0.29	2.6	0.0
	5	0.9	1.1	0.9	0.22	0.6	0.2	0.9	1.1	0.9	0.22	0.8	0.0
	9	0.9	1.0	1.5	0.07	0.3	0.1	0.9	1.0	1.6	0.05	0.3	0.0
Hx $\rightarrow$ Pen	2	0.5	1.6	−2.2	0.95	28.1	1.2	0.4	1.7	−2.1	0.95	16.7	1.0
	3	0.8	1.3	−0.2	0.61	4.8	0.5	0.8	1.3	−0.3	0.70	2.5	0.1
	4	0.9	1.1	0.8	0.25	1.1	0.0	0.9	1.2	0.9	0.14	0.9	0.0
	5	0.9	1.1	0.9	0.22	0.6	0.1	0.9	1.1	0.8	0.17	0.9	0.2
	9	0.9	1.0	1.5	0.09	0.4	0.1	0.9	1.0	1.6	0.06	0.3	0.0
Hep $\rightarrow$ Hx	2	0.5	1.6	−2.1	0.97	24.8	4.2	0.5	1.6	−1.9	0.82	18.7	2.9
	3	0.8	1.3	−0.2	0.65	6.4	0.3	0.8	1.3	−0.3	0.37	4.3	0.4
	4	0.9	1.2	1.0	0.12	1.0	0.0	0.9	1.1	0.9	0.23	0.6	0.0
	5	0.9	1.1	0.9	0.46	2.2	0.1	0.9	1.1	0.8	0.13	1.1	0.2
	9	0.9	1.0	1.4	0.08	0.5	0.0	0.9	1.0	1.6	0.12	0.6	0.1

^aThe overlap measures are described in the main text. Values that violate the new criteria ($\Omega > 0.85$, $|K_{AB} - 1| < 0.25$, $\Pi > 0.5$, $w_{max} < 0.5$, $\Delta\Delta G_{EA} < 2$ kJ/mol, and $\Delta\Delta G_{TI} < 0.25$ kJ/mol) are marked in bold type.

calculations gave a diverging conclusion. For $\Delta\Delta G_{EA}$, all of the values between 1.8 and 3.2 kJ/mol gave three discrepancies compared with the Π measure. $\Delta\Delta G_{TI}$ seemed to be the most problematic and least useful criterion. Clearly, 4 kJ/mol was much too large, but all of the values between 0.21 and 0.29 kJ/mol gave six discrepancies compared with the Π measure. This is probably the case because these low levels are close to the precision of the estimates and they are also affected by integration errors in TI. Therefore, we do not recommend $\Delta\Delta G_{TI}$ as an overlap criterion. For the other measures, we suggest the following improved criteria for proper overlap: $\Omega > 0.85$, $|K_{AB} - 1| < 0.25$, $\Pi > 0.5$, $w_{max} < 0.5$, and $\Delta\Delta G_{EA} < 2$ kJ/mol. Values violating these criteria are shown in bold face in Table 2. Future studies on other systems are needed to see how general these criteria are.

Finally, we combined the $\Delta\Delta G_{L_i}^{MM \rightarrow QM/MM}$ estimates with the MM estimates of the binding free energies ($\Delta\Delta G_{L_0 \rightarrow L_1}^{MM}$ in eq 3)²⁸ to get our final QM/MM estimates of the binding affinities. These results are collected in Table 3. The MM $\rightarrow$ QM corrections for the relative binding affinities (with nine Λ values; $\Delta\Delta G_{L_i}^{MM \rightarrow QM/MM} - \Delta\Delta G_{L_0}^{MM \rightarrow QM/MM}$) were rather small (−4 to +5 kJ/mol), but the estimated relative affinities are also rather small (2–13 kJ/mol), so the effect of the correction is quite significant.

As for the MM $\rightarrow$ QM/MM corrections, the final $\Delta\Delta G_{bind}$ estimates obtained with two Λ values were highly uncertain and inaccurate, with standard errors of 3–8 kJ/mol. With more Λ values, the precision improved from 1.2–1.4 kJ/mol (three Λ) to 0.7–1.0 kJ/mol (nine Λ). The results with three, four, and five Λ

Table 3. Final QM/MM Estimates of the Relative Binding Affinities (in kJ/mol) Obtained by Reference-Potential Calculations with Two to Nine Λ Values and by Direct QM/MM-FEP Calculations (Column “horiz.”); For Comparison, Experimental Affinities⁴⁹ and Our Results Obtained at the MM Level²⁸ Are Also Included, Along with MAD, R^2 , and τ_r Measures of the Various Sets of Estimates Compared with the Experimental Affinities

	MM	two Λ values	three Λ values	four Λ values	five Λ values	nine Λ values	horiz.	exptl
MeBz $\rightarrow$ Bz	15.9 $\pm$ 0.1	12.5 $\pm$ 6.8	16.8 $\pm$ 1.4	16.3 $\pm$ 0.5	16.4 $\pm$ 0.9	16.3 $\pm$ 0.7		8.9 $\pm$ 0.5
EtBz $\rightarrow$ MeBz	1.0 $\pm$ 0.1	2.7 $\pm$ 5.9	0.0 $\pm$ 1.4	−1.2 $\pm$ 0.5	−1.7 $\pm$ 0.9	−1.9 $\pm$ 0.7		1.7 $\pm$ 0.5
pClBz $\rightarrow$ Bz	19.1 $\pm$ 0.1	18.0 $\pm$ 7.6	22.8 $\pm$ 1.3	23.8 $\pm$ 0.5	22.7 $\pm$ 0.9	23.7 $\pm$ 0.7	23.2 $\pm$ 0.9	12.5 $\pm$ 0.2
mClBz $\rightarrow$ Bz	7.1 $\pm$ 0.1	17.4 $\pm$ 5.8	9.9 $\pm$ 1.4	10.3 $\pm$ 0.6	10.2 $\pm$ 0.9	9.9 $\pm$ 0.7	11.3 $\pm$ 0.8	6.4 $\pm$ 0.3
Hx $\rightarrow$ Bz	13.1 $\pm$ 0.3	11.1 $\pm$ 3.4	9.9 $\pm$ 1.3	9.5 $\pm$ 0.6	9.0 $\pm$ 0.9	9.4 $\pm$ 0.7		7.9 $\pm$ 0.4
MeHx $\rightarrow$ Hx	15.4 $\pm$ 0.2	21.9 $\pm$ 3.4	18.5 $\pm$ 1.2	18.1 $\pm$ 0.6	19.4 $\pm$ 0.9	18.7 $\pm$ 0.6		8.3 $\pm$ 0.4
Hx $\rightarrow$ Pen	7.5 $\pm$ 0.7	4.1 $\pm$ 4.0	6.5 $\pm$ 1.4	7.9 $\pm$ 0.9	6.3 $\pm$ 1.1	7.8 $\pm$ 1.0		7.9 $\pm$ 0.4
Hep $\rightarrow$ Hx	5.5 $\pm$ 0.7	16.5 $\pm$ 3.9	7.1 $\pm$ 1.4	7.0 $\pm$ 0.8	7.9 $\pm$ 1.1	7.5 $\pm$ 0.9		4.1 $\pm$ 0.3
MAD	3.6 $\pm$ 0.2	6.6 $\pm$ 1.5	4.9 $\pm$ 0.5	4.7 $\pm$ 0.2	5.2 $\pm$ 0.4	5.0 $\pm$ 0.3		
R^2	0.84 $\pm$ 0.04	0.21 $\pm$ 0.16	0.79 $\pm$ 0.07	0.86 $\pm$ 0.04	0.76 $\pm$ 0.06	0.83 $\pm$ 0.05		
τ_r	1.00 $\pm$ 0.00	1.00 $\pm$ 0.15	0.75 $\pm$ 0.13	0.75 $\pm$ 0.01	0.75 $\pm$ 0.04	0.75 $\pm$ 0.01		

values agreed with those obtained with nine Λ values within 1.9, 0.5, and 0.8 kJ/mol (0.7, 0.5, and 0.6 kJ/mol on average), respectively.

The calculated relative affinities can be compared with experimental affinities,⁴⁹ which are also included in Table 3. The results were quite good, giving (with nine Λ values) a mean absolute deviation (MAD) of 5.0 $\pm$ 0.3 kJ/mol, a correlation coefficient (R^2) of 0.83 $\pm$ 0.5, and the correct sign for all of the affinities except the smallest one ($\tau_r = 0.75 \pm 0.01$). The results with only two Λ values were much worse, with $R^2 = 0.21 \pm 0.16$ and MAD = 6.6 $\pm$ 1.5 kJ/mol. However, the results with three to nine Λ values were statistically equivalent ($R^2 = 0.76\text{--}0.86$ and MAD = 4.7–5.2 kJ/mol).

The present results are somewhat better than our previous QM/MM FEP results obtained with ssEA, which gave no correlation ($R^2 = 0.00 \pm 0.03$) but a similar MAD of 4.9 $\pm$ 0.4 kJ/mol.²⁹ However, the results are still not better than the original MM results:²⁸ the correlation coefficient is similar ($R^2 = 0.83$ compared to 0.84), but the MAD is slightly worse (5.0 $\pm$ 0.3 kJ/mol compared with 3.6 $\pm$ 0.2 kJ/mol). In fact, as can be seen in Figure 3, the QM/MM calculations improved the free energies of only two of the eight transformations.

Direct QM/MM FEP. For comparison, we also performed some full QM/MM FEP calculations, i.e., by performing all of the

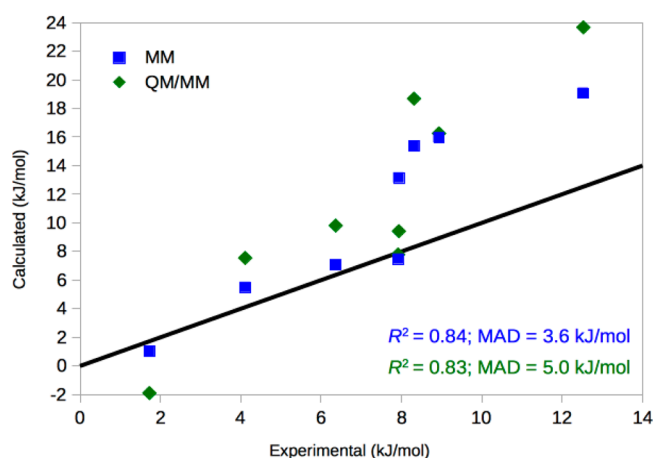


Figure 3. Comparison of the experimental and calculated affinities obtained with either MM or the reference-potential QM/MM method with nine Λ values. The line shows the perfect correlation.

MD simulations at the QM/MM level and calculating the QM/MM free energy difference $\Delta\Delta G_{\text{bind}}$ directly using the upper horizontal route in Figure 2 (i.e., using no MM simulations). For technical reasons, these calculations could only be performed for the two Cl $\rightarrow$ H perturbations (i.e., the pClBz $\rightarrow$ Bz and mClBz $\rightarrow$ Bz perturbations). The overlap measures indicated problems for many of the calculations with large λ values ($\lambda = 0.6\text{--}1.0$). This was solved by adding simulations at four or five additional intermediate λ values (0.65, 0.75, 0.85, 0.93, and 0.98).

The results of these calculations are also included in Table 3. It can be seen that the results agree reasonably well with the calculations with the reference potential: For the pClBz $\rightarrow$ Bz perturbation, $\Delta\Delta G_{\text{bind}} = 23.2 \pm 0.9$ kJ/mol, compared to 23.7 ± 0.7 kJ/mol with the reference potential and nine Λ values. For the mClBz $\rightarrow$ Bz perturbation, the two results were 11.3 ± 0.8 and 9.9 ± 0.7 kJ/mol. This indicates that the two sets of results were consistent and that the calculations were converged.

CONCLUSIONS

In this paper, we have compared two approaches to obtain relative binding free energies at the QM/MM level, viz., by direct QM/MM FEP calculations (upper horizontal route in Figure 2) and with the reference-potential approach (the other three routes in Figure 2). In variance to most previous approaches,^{28–31,33} we have improved the convergence of the MM $\rightarrow$ QM/MM perturbations by actually performing QM/MM MD simulations for a number Λ values. We have shown that the two approaches give consistent results within the statistical uncertainty, provided that the convergence of both approaches is carefully monitored and that proper overlap between states is ensured.

In particular, we have studied how many Λ values are needed for the reference-potential calculations. The results in Tables 1 and S1 clearly show that two Λ values are not enough to obtain any reliable results, giving errors of 2–9 kJ/mol in the final free energies and a statistical precision of 3–8 kJ/mol. TI and EA calculations gave even larger errors with few Λ values. This shows that a linear-response approximation (i.e., the TI results with two Λ values) is inappropriate for the present systems. This approximation is often used in the paradynamics approach without any test of the overlap.^{26,42–44} In fact, our results strongly suggest that MBAR or BAR calculations should be employed for this type of calculation, where the overlap is often problematic.

Already with three Λ values the results are much improved, showing errors of less than 2 kJ/mol (0.7 kJ/mol on average)

compared to those with nine Λ values and a precision of 1.2–1.4 kJ/mol. However, a number of overlap entries indicate that it may be dangerous to trust these results (cf. Table 2). Instead, we tend to prefer the results obtained with four Λ values, for which the overlap is proper, the maximum and average errors are reduced to 0.6 and 0.3 kJ/mol, and the precision is 0.5–0.9 kJ/mol (with a sampling frequency of 1 ps).

We strongly emphasize the importance of **monitoring the overlap for the MM $\rightarrow$ QM/MM perturbations**. We have tested six different overlap criteria: Ω , K_{AB} , Π , $w_{\max}$, $\Delta\Delta G_{EA}$, and $\Delta\Delta G_{TI}$. In accordance with our previous studies,¹⁶ Π seemed to be the most reliable measure, whereas $\Delta\Delta G_{TI}$ cannot be recommended. The results were so consistent that they could be used to suggest a new set of stricter criteria for proper overlap: $\Omega > 0.85$, $|K_{AB} - 1| < 0.25$, $\Pi > 0.5$, $w_{\max} < 0.5$, and $\Delta\Delta G_{EA} < 2$ kJ/mol. Moreover, the MBAR precision also provides a useful guide whether the calculations are reliable. It is possible that these values may depend on the simulated system, and the same most likely also applies to the number of Λ values needed to obtain converged results. Therefore, great caution is recommended when studying a new system, carefully following the convergence of the free energies with the help of the overlap measures and comparing the results obtained with different numbers of Λ values.

Most importantly, the present calculations allow us to compare the efficiencies of various approaches to calculate binding free energies at the QM/MM level. The reference-potential approach with four Λ values required in total 12 QM/MM MD simulations (three simulations each for the two ligands, both when bound to the host and when free in solution; the $\Lambda = 0$ simulation can be performed at the MM level). The direct QM/MM FEP instead required 34–36 λ values because of the **poor overlap observed at high λ values**. This indicates that the reference-potential approach actually is **~ 3 times more effective** than the direct approach. Moreover, it seems likely that the horizontal perturbations may require longer simulations for convergence than the vertical ones in Figure 2 because the latter involve only a single ligand, which normally binds in a single conformation, whereas the binding mode may change when the ligands are changed. In addition, the computational effort of the reference-potential approach can be further reduced by observing that the MM $\rightarrow$ QM/MM perturbation is performed for a single ligand, which shows that if the same ligand is involved in several perturbations, **only the MM FEP needs to be redone, not the MM $\rightarrow$ QM/MM step for the common ligand**. In the present case, this means that we could have omitted one MM $\rightarrow$ QM/MM perturbation for MeBz as well as three each for the Bz and Hx ligands, i.e., avoiding seven out of 16 MM $\rightarrow$ QM/MM perturbations. Thus, in total the reference-potential approach would be over 5 times more effective for the present test set.

The reference-potential method should also be compared to the ssEA approach used in our previous study.²⁹ We have used a total simulation time of **1.5 ns** for each of the 12 MM $\rightarrow$ QM/MM perturbations in this study, each requiring 1.5 million QM/MM energy evaluations (with a time step of **1 fs**). Thus, the present approach is ~ 25 times more time-consuming than the ssEA approach used in our previous study.²⁹ On the other hand, the previous study involved at least two approximations with potentially serious influence on the results. First, it was essential to employ the cumulant approximation to calculate the ssEA MM $\rightarrow$ QM/MM free energies. This requires that the energy distribution is Gaussian, which is hard to prove for the low-energy tail that determines the free energy difference. In fact,

even if the bootstrapped standard errors indicated that the free energies were converged to within 1 kJ/mol, the individual $\Delta\Delta G_{L}^{s,MM \rightarrow QM/MM}$ terms still showed differences of up to 9 kJ/mol when taken from independent simulations. In the present study, the corresponding terms were converged to better than 1 kJ/mol. Second, the previous study employed only interaction energies ($E(\text{complex}) - E(\text{host}) - E(\text{ligand})$) in the MM $\rightarrow$ QM/MM calculations in order to improve the convergence. This excludes differences in the internal energies of the host and the ligand and may affect the results in an uncontrolled manner.³³ Finally, the present approach is fully automatic, employing only standard QM/MM software, whereas the previous approach required separate QM calculations of tens of thousands of structures, giving very many files and much overhead. For all of these reasons, we tend to prefer the present reference-potential approach.

Even if the present results are the best QM/MM results obtained for this system, it is of course somewhat disappointing that we still do not get any consistent improvement over the original MM calculations. However, this was not the aim of the present study. Instead, the QM method and size of the QM system (PM6-DH+ only for the ligand) were selected to allow for long and converged calculations and testing of several different approaches. If the prime aim is to improve the results, a more accurate method (e.g., dispersion-corrected DFT) should be used for the ligand as well as the host and the closest water molecules. The present study shows how such an investigation should be performed and allows for an estimate of the time consumption.

效果并不好，重在思路。。。

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jctc.6b01217.

The $\Delta G_{L}^{\text{bound,MM} \rightarrow \text{QM/MM}}$ and $\Delta G_{L}^{\text{free,MM} \rightarrow \text{QM/MM}}$ QM/MM free energy corrections for the eight transformations (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: Ulf.Ryde@teokem.lu.se. Tel: +46–46 2224502. Fax: +46–46 2228648.

ORCID

Ulf Ryde: 0000-0001-7653-8489

Funding

This investigation was supported by grants from the Swedish Research Council (Project 2014-5540) and the Knut and Alice Wallenberg Foundation (KAW 2013.0022). The computations were performed on computer resources provided by the Swedish National Infrastructure for Computing (SNIC) at Lunarc at Lund University and HPC2N at Umeå University.

Notes

The authors declare no competing financial interest.

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